

PROJECT NAME: EMPLOYEE MANAGEMENT SYSTEM

1. C++ SOURCE CODE

```
#include <iostream>
#include <fstream>
#include <cstring>
#include <iomanip>

using namespace std;

class Employee {
private:
    int id;
    char name[40];
    char department[30]; // Added for Department-Wise categorization
    char designation[30];
    double baseSalary;

    // Payroll and Attendance fields
    double netSalary;
    int attendanceCount; // Added for Attendance tracking

public:
    void calculatePayroll() {
        double hra = baseSalary * 0.15; // 15% House Rent Allowance
        double da = baseSalary * 0.10; // 10% Dearness Allowance
        double pf = baseSalary * 0.12; // 12% Provident Fund Deduction
        netSalary = (baseSalary + hra + da) - pf;
    }

    void getData() {
        cout << "\nEnter Employee ID (Integer): ";
        cin >> id;
        cin.ignore();
        cout << "Enter Name: ";
        cin.getline(name, 40);

        // Department Input Option Menu
        cout << "Select Department:\n1. ECE\n2. IT\n3. HR\n4. Finance\nEnter choice (1-4): ";
        int deptChoice;
        cin >> deptChoice;
        cin.ignore();
        switch(deptChoice) {
            case 1: strcpy(department, "ECE"); break;
            case 2: strcpy(department, "IT"); break;
            case 3: strcpy(department, "HR"); break;
            case 4: strcpy(department, "Finance"); break;
            default: strcpy(department, "General"); break;
        }

        cout << "Enter Designation: ";
        cin.getline(designation, 30);
        cout << "Enter Base Salary (₹): ";
        cin >> baseSalary;
    }
};
```

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        // Initialize attendance to 0 for a new record
        attendanceCount = 0;
        calculatePayroll();
    }

    void markAttendance() {
        attendanceCount++;
        cout << "\nAttendance marked successfully! Current active days: " <<
attendanceCount << endl;
    }

    void showData() const {
        cout << left << setw(6) << id
            << setw(18) << name
            << setw(12) << department
            << setw(15) << designation
            << setw(12) << netSalary
            << setw(10) << attendanceCount << endl;
    }

    int getId() const { return id; }
    const char* getDepartment() const { return department; }
};

// Function prototypes for handling system loops
bool login();
void addEmployee();
void displayDepartmentWise();
void searchEmployee();
void markEmployeeAttendance();

int main() {
    cout << "=====\n";
    cout << "  EMPLOYEE MANAGEMENT SYSTEM (DEPT-WISE & ATTENDANCE)  \n";
    cout << "=====\n";

    if (!login()) {
        cout << "\nAccess Denied. Program exiting.\n";
        return 0;
    }

    int choice;
    while (true) {
        cout << "\n--- SYSTEM CONTROL MENU ---\n";
        cout << "1. Register New Employee\n";
        cout << "2. Display Records (Department-Wise)\n";
        cout << "3. Search Employee Profile by ID\n";
        cout << "4. Mark Daily Attendance\n";
        cout << "5. Exit Application\n";
        cout << "Enter administrative option: ";
        cin >> choice;

        switch (choice) {
            case 1: addEmployee(); break;
            case 2: displayDepartmentWise(); break;
            case 3: searchEmployee(); break;
            case 4: markEmployeeAttendance(); break;
            case 5:
                cout << "\nSystem closing. All modifications saved to storage.\n";

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        return 0;
    default:
        cout << "\nSelection out of range! Choose options between 1 and 5.\n";
    }
}
return 0;
}

bool login() {
    string password;
    int attempts = 0;
    while (attempts < 3) {
        cout << "\nEnter Admin Access Password: ";
        cin >> password;
        if (password == "admin123") {
            cout << "\nAuthentication Successful!\n";
            return true;
        } else {
            attempts++;
            cout << "Invalid Passkey! Security attempts remaining: " << (3 - attempts)
<< "\n";
        }
    }
    return false;
}

void addEmployee() {
    Employee emp;
    emp.getData();

    ofstream outFile("records.dat", ios::app | ios::binary);
    if (!outFile) {
        cout << "\nError accessing storage stream file!\n";
        return;
    }
    outFile.write(reinterpret_cast<char*>(&emp), sizeof(Employee));
    outFile.close();
    cout << "\nProfile committed onto local physical sectors successfully!\n";
}

void displayDepartmentWise() {
    Employee emp;
    ifstream inFile("records.dat", ios::binary);
    if (!inFile) {
        cout << "\nNo records available to process.\n";
        return;
    }

    cout << "\nSelect Department to display:\n1. ECE\n2. IT\n3. HR\n4. Finance\nEnter
choice: ";
    int choice;
    cin >> choice;
    char targetDept[30];
    switch(choice) {
        case 1: strcpy(targetDept, "ECE"); break;
        case 2: strcpy(targetDept, "IT"); break;
        case 3: strcpy(targetDept, "HR"); break;
        case 4: strcpy(targetDept, "Finance"); break;
        default: strcpy(targetDept, "General"); break;
    }
}

```

```

        cout <<
"\n=====\\n";
        cout << left << setw(6) << "ID" << setw(18) << "Name" << setw(12) << "Dept" <<
setw(15) << "Designation" << setw(12) << "Net Pay(₹)" << setw(10) << "Attendance" <<
endl;
        cout <<
"=====\\n";

        bool found = false;
        while (inFile.read(reinterpret_cast<char*>(&emp), sizeof(Employee))) {
            if (strcmp(emp.getDepartment(), targetDept) == 0) {
                emp.showData();
                found = true;
            }
        }
        if (!found) {
            cout << "          No employees found registered under department: " << targetDept
<< endl;
        }
        cout <<
"=====\\n";
        inFile.close();
    }

void searchEmployee() {
    Employee emp;
    ifstream inFile("records.dat", ios::binary);
    if (!inFile) {
        cout << "\\nStorage system is blank.\\n";
        return;
    }

    int searchId;
    bool found = false;
    cout << "\\nEnter Target Employee ID to query: ";
    cin >> searchId;

    while (inFile.read(reinterpret_cast<char*>(&emp), sizeof(Employee))) {
        if (emp.getId() == searchId) {
            cout << "\\nMatch Located:\\n";
            cout <<
"-----\\n";
            emp.showData();
            cout <<
"-----\\n";
            found = true;
            break;
        }
    }
    if (!found) cout << "\\nEmployee target key " << searchId << " does not match index
fields.\\n";
    inFile.close();
}

void markEmployeeAttendance() {
    Employee emp;
    fstream file("records.dat", ios::in | ios::out | ios::binary);
    if (!file) {
        cout << "\\nStorage missing or inaccessible.\\n";
    }
}

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        return;
    }

    int targetId;
    bool found = false;
    cout << "\nEnter Employee ID to log today's attendance: ";
    cin >> targetId;

    // Cycle through binary blocks to find match and write shift updates
    while (file.read(reinterpret_cast<char*>(&emp), sizeof(Employee))) {
        if (emp.getId() == targetId) {
            emp.markAttendance();

            // Back up stream pointer position by size of one object block to overwrite
            // modifications
            int pos = static_cast<int>(file.tellg()) - sizeof(Employee);
            file.seekp(pos);
            file.write(reinterpret_cast<char*>(&emp), sizeof(Employee));
            found = true;
            break;
        }
    }
    if (!found) cout << "\nEmployee matching ID " << targetId << " not found.\n";
    file.close();
}

```

2. PROGRAM OUTPUT (TERMINAL SIMULATION)

```

=====
EMPLOYEE MANAGEMENT SYSTEM (DEPT-WISE & ATTENDANCE)
=====

Enter Admin Access Password: admin123

Authentication Successful!

--- SYSTEM CONTROL MENU ---
1. Register New Employee
2. Display Records (Department-Wise)
3. Search Employee Profile by ID
4. Mark Daily Attendance
5. Exit Application
Enter administrative option: 1

Enter Employee ID (Integer): 2048
Enter Name: Dhruv
Select Department:
1. ECE
2. IT
3. HR
4. Finance
Enter choice (1-4): 1
Enter Designation: Lead Engineer
Enter Base Salary (₹): 75000

Profile committed onto local physical sectors successfully!

```

```

--- SYSTEM CONTROL MENU ---
1. Register New Employee
2. Display Records (Department-Wise)
3. Search Employee Profile by ID
4. Mark Daily Attendance
5. Exit Application
Enter administrative option: 4

Enter Employee ID to log today's attendance: 2048

Attendance marked successfully! Current active days: 1

--- SYSTEM CONTROL MENU ---
1. Register New Employee
2. Display Records (Department-Wise)
3. Search Employee Profile by ID
4. Mark Daily Attendance
5. Exit Application
Enter administrative option: 2

Select Department to display:
1. ECE
2. IT
3. HR
4. Finance
Enter choice: 1

=====
ID      Name           Dept      Designation  Net Pay(₹)  Attendance
=====
2048   Dhruv              ECE       Lead Engineer  84750       1
=====

--- SYSTEM CONTROL MENU ---
1. Register New Employee
2. Display Records (Department-Wise)
3. Search Employee Profile by ID
4. Mark Daily Attendance
5. Exit Application
Enter administrative option: 3

Enter Target Employee ID to query: 2048

Match Located:
-----
2048   Dhruv              ECE       Lead Engineer  84750       1
-----

--- SYSTEM CONTROL MENU ---
1. Register New Employee
2. Display Records (Department-Wise)
3. Search Employee Profile by ID
4. Mark Daily Attendance
5. Exit Application
Enter administrative option: 5

System closing. All modifications saved to storage.

```